

EMSLD : Efficient Multiple Strip Large-Kernel Decocder for Medical Segmentation

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Abstract—While Vision Mamba has emerged as a formidable alternative to Transformers for global context modeling, its efficacy in decoding complex medical structures remains limited. This paper introduces Strip Large-kernel Mamba for Medical Image Segmentation (SLMambaMedseg), a novel hybrid architecture designed to bridge the gap between long-range dependency and directional feature refinement. Our core innovation lies in the Multi-scale Strip Large-kernel Convolutional Block (MSLCB), which replaces conventional decoders to explicitly capture the anisotropic features of slender organs. By integrating a Strip Large-kernel Gated Attention (SLGA) mechanism, the model effectively filters multi-scale noise while amplifying critical anatomical details. Extensive evaluations on Synapse and ACDC benchmarks demonstrate that SLMambaMedseg outperforms state-of-the-art models, particularly in small-object segmentation, achieving a DSC of 83.36% and an impressive HD95 of 16.18 mm on Synapse.

I. INTRODUCTION

Automated segmentation of anatomical structures in medical images is foundational for computer-aided diagnosis and treatment planning[1], [2], [3]. While Convolutional Neural Networks (CNNs) long dominated this field with U-Net, their inherent limitation in modeling long-range dependencies has motivated the shift to self-attention mechanisms[4], [5]. However, the quadratic complexity of Transformers imposes significant computational burden. Vision Mamba[6], [7], [8], built on selective state space models (SSMs), has recently emerged as a compelling alternative, achieving global context modeling with linear complexity.

Vision Mamba has been integrated as an encoder in segmentation frameworks[9], [10]. Yet a critical gap persists: Mamba’s strength lies in modeling rather than generation, so existing models predominantly pair it with separate decoders[11], [12], leaving the full potential of a unified Mamba-centric architecture untapped.

To bridge this gap, we propose SLMambaMedseg, a hybrid architecture integrating a Vision Mamba encoder with a comprehensively enhanced convolutional decoder.

The encoder employs hierarchical VSS blocks for efficient global representation. The decoder introduces a Multi-scale Strip Large-kernel Convolutional Block (SLCB) and a Strip Large-Kernel Grouped Attention Gate (SLGAG). Inspired by large-kernel CNNs[13] and strip convolutions[14], SLCB uses parallel strip and large-kernel depthwise separable convolutions to capture anisotropic features and broad context simultaneously. SLGAG refines skip-connection fusion, preserving spatial details during upsampling.

The principal contributions are:

- 1) We introduce SLMambaMedseg that synergistically combines a Vision Mamba encoder with a heavily augmented convolutional decoder.
- 2) We design SLCB and SLGAG, which systematically incorporate strip and large-kernel convolutions to address the geometric characteristics of medical organs.
- 3) Extensive experiments on Synapse and ACDC demonstrate state-of-the-art performance with markedly improved efficiency.

II. RELATED WORKS

A. Medical image segmentation

Medical image segmentation has been driven by the trade-off between local feature extraction and global context modeling. U-Net[1], built on CNNs, established the de facto standard, but the intrinsic locality of convolution restricts long-range dependency modeling[16], [15], limiting performance on structures with large scale variations[4].

Vision Transformers (ViTs)[17] addressed this by adopting self-attention for global context. TransUNet[4], Swin-UNet[5], and SegFormer[18] have demonstrated strong results in medical segmentation using Transformer-based designs.

However, the quadratic complexity of self-attention imposes heavy computational burden[19], spurring interest in hybrid architectures that balance efficiency and performance[20]. TransUNet[4] exemplifies this paradigm by combining a CNN encoder with Transformer bottleneck. Our SLMambaMedseg follows this hybrid philosophy, further replacing the Transformer with the more efficient Vision Mamba.

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B. State Space Models

State Space Models (SSMs) capture long-range dependencies with linear complexity, contrasting sharply with the quadratic scaling of self-attention[6], [19]. The Structured State Space Sequence model (S4)[22] integrated the HiPPO framework[21] for long-range memory, enabling efficient parallel training. However, S4's time-invariant parameters remain independent of input data[23], limiting context-dependent reasoning critical for dense visual tasks.

Mamba[6] overcame this by introducing input-dependent transition matrices, enabling selective information propagation. This allows Mamba-based architectures to match Transformer-level representation while retaining linear complexity, making them a compelling alternative for medical vision tasks[8], [7], [9].

III. METHODOLOGY

This section will first introduce the overall architecture of SLMambaMedseg, followed by a detailed description of each individual module used within it, including various technical principles, implementation details, and parameter settings.

A. SLMambaMedseg

As depicted in Figure 1 (a), the overall architecture of SLMambaMedseg is presented, which is based on U-Shape-Network[1]. Specifically, SLMambaMedseg comprises a Patch Embedding layer as initial processing, four-stage VSS Blocks[7], [6] as the encoder, four-stage SLCBs as the decoder, a SH layer as final projection and SLGAGs as skip connections.

Firstly, in the Patch Embedding layer input image is divided from $x = R^{H \times W \times 3}$ into $x' \in R^{\frac{H}{4} \times \frac{W}{4} \times C}$, each patch is non-overlapping and the size is 4×4 , subsequently mapping the dimensions of the image to C. Before entering the encoder layers, x' will be normalized.

For the encoder, eight VSS Blocks are employed distributed across four stages, with each stage comprising two VSS Blocks. A patch merging operation is applied at the end of the first three stages to downsample the spatial dimensions of the input features while simultaneously doubling the channel capacity [24]. Consequently, the channel counts for the four stages are denoted as [C, 2C, 4C, 8C].

Similarly, the decoder utilizes four Multi-scale Strip Large-kernel Convolutional Blocks (SLCBs) to progressively refine the hierarchical pyramid features extracted from the encoder via skip connections (i.e., x_1 , x_2 , x_3 , and x_4 , as illustrated in Figure 1) [25]. Furthermore, a patch expanding operation is employed following the initial three SLCBs to progressively restore the spatial

resolution [5]. The channel counts for the decoder stages follow the reverse order: [8C, 4C, 2C, C].

To bridge the semantic gap between the encoder and decoder, three Strip Large-Kernel Grouped Attention Gates (SLGAGs) are integrated within the skip connections [26]. The features that are upsampled in the patch expanding layer are subsequently element-wise added to the refined outputs from the corresponding SLGAGs.

Finally, a Segmentation Head (SH) layer is applied to project the fused features and produce the final segmentation output.

B. VSS Block

As depicted in Figure 1 (b), the VSS Block, derived from VMamba[8], [7], is the core module of our encoder. Firstly, the input patches x are layer normalized, then split into two branches, where one branch performs global modeling but loses local features, and the other branch preserves local features.

The front branch firstly processes the input patches through a linear layer $LL(\cdot)$ and a depthwise separable convolution $DWC(\cdot)$, with an activation function $R(\cdot)$ at the end. We then feed the patches into $SS2D(\cdot)$ for future feature extraction. Finally, the features are layer normalized by $LN(\cdot)$ and await output patches from the other branch.

As the front branch loses local features through global modeling, the local features will be preserved in the other branch, where patches will only undergo a linear layer $LL(\cdot)$ and an activation function $R(\cdot)$.

The outputs from both branches then undergo a bit-wise multiplication (Hadamard Product) after passing through another linear layer $LL(\cdot)$. Finally, the merged features are added bit-wise to the initial input x .

The $VSS(\cdot)$ can be formulated as in Equation 1:

$$VSS(x) = x + SS2D(LN(DWC(LL(x)))) \otimes \sigma(LL(x)) \quad (1)$$

C. Strip Large-kernel grouped attention gate (SLGAG)

We introduce a new strip large-kernel group attention gate (SLGAG) to enhance the activation of relevant features while suppressing irrelevant ones.

As shown in Figure 1 (c), SLGAG accepts two inputs, g and x , from a patch expanding layer in the decoder and a patch merging layer that is one level higher than the patch expanding layer; both inputs have the same size to ensure bit-wise addition operates correctly. SLGAG processes g and x using a 3×3 convolution ($Cnv_{3 \times 3}(\cdot)$) and a strip convolution ($SC(\cdot)$) to extract features. These convolved features are then batch-normalized using $BN(\cdot)$ and merged via bit-wise addition. The merged feature map is activated by a ReLU layer $R(\cdot)$. Afterward, a 1×1 convolution ($Cnv_{1 \times 1}(\cdot)$) is applied to obtain a single-channel feature map. Then the single-channel feature map will pass through a $BN(\cdot)$ and

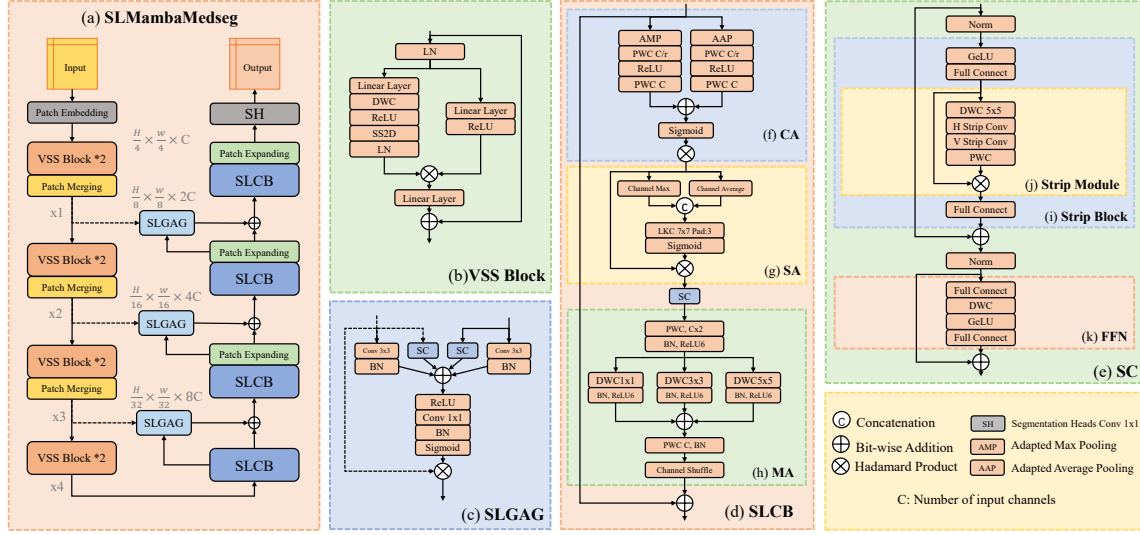


Fig. 1. (a) The overall architecture of SLMambaMedseg. (b) VSS Block is the main construction of our encoder. (c) Strip Large-kernel grouped attention gate (SLGAG) can enhance features from skip connections. (d) Multi-scale Strip Large-kernel convolutional block (SLCB), progressing patches in all dimensions to further enhance features, is the main component of our decoder. (e) Strip convolutional block (SC). (f) Channel attention block (CA). (g) Spatial attention block (SA). (h) Multi-scale attention block (MA). (i) Strip block. (j) Strip module. (k) Feed-Forward Network (FFN).

an activation function $\sigma(\cdot)$ to successfully yield the attention coefficients.

Finally, initial input x and the activated single-channel feature map enter a bit-wise multiplication to produce the attention-gated feature.

The $SLGAG(\cdot)$ can be formulated as in Equation 2:

$$SLGAG(g, x) = x \otimes \sigma(BN(Cnv_{1 \times 1}(R(SC(g) + BN(Cnv_{3 \times 3}(g)) + SC(x) + BN(Cnv_{3 \times 3}(x)))))) \quad (2)$$

D. Multi-scale Strip Large-kernel convolutional block (SLCB)

We introduce a new multi-scale strip large-kernel convolutional block as depicted in Figure 1 (d). Our SLCB contains a channel attention block (CA) that finds out most relevant channel, a spatial attention block (SA) that find out the most relevant feature maps, a strip convolution block (SC) that enhances feature extraction capacity for tiny details, and a multi-scale attention block (MA) that enhances contextual relationships among all of the features.

Channel Attention Block (CA)[27]: As depicted in Figure 1 (f), the input patches x first enter the channel attention (CA) module and are then split into two branches, where x undergo adaptive max pooling ($AMP(\cdot)$) and adaptive average pooling ($AAP(\cdot)$) respectively, to extract the most significant feature per channel from the entire feature map. For each pooled feature map, two point-wise convolution blocks are applied: $PWC_{C/r}(\cdot)$ (with $r = 16$) to reduce the number

of channels and $PWC_C(\cdot)$ to recover it, with an activation function $R(\cdot)$ in between. The two pooled feature maps are then combined via bit-wise addition before passing through an activation function $\sigma(\cdot)$. Finally, a bit-wise multiplication is applied between the combined feature map and the initial input x to get the final output. The $CA(\cdot)$ can be formulated as in Equation 3:

$$CA(x) = x \otimes \sigma(PWC_C(R(PWC_{C/r}((AMP(x)))) + PWC_C(R(PWC_{C/r}(AAP(x)))))) \quad (3)$$

Spatial Attention Block (SA)[27]: As depicted in Figure 1 (g), following a design similar to that of the CA module, the input patches x first undergo max pooling ($CH_{max}(\cdot)$) and average pooling ($CH_{average}(\cdot)$) in two separate branches along the channel dimension to capture local features. The outputs are then concatenated before being fed into a large-kernel convolution layer $LKC(\cdot)$ (i.e., kernel size is 7×7 and padding is 3), where local contextual relationships among the features are enhanced. Finally, the concatenated feature maps are passed through an activation function $\sigma(\cdot)$ and bit-wise multiplied (Hadamard product) with the initial input x .

The $SA(\cdot)$ can be formulated as in Equation 4:

$$SA(x) = x \otimes \sigma(LKC([CH_{max}(x); CH_{average}(x)])) \quad (4)$$

Multi-scale Attention Block (MA): As depicted in Figure 1 (h), the input patches x first undergo channel expansion via a point-wise convolution $PWC_{C \times 2}(\cdot)$ (with an expansion factor of 2), followed by batch normalization $BN(\cdot)$ and an activation function $R6(\cdot)$

(ReLU6). Then, the patches are separately passed through three depthwise separable convolution layers ($DWC(\cdot)$) with different kernel sizes to capture multi-scale and multi-resolution contexts (the number and size of the kernels are configurable). After the features from different channels are summed, a channel shuffle layer $CSF(\cdot)$ will be applied to enhance cross-channel relationships. The shuffled feature maps are then passed through another point-wise convolution $PWC_C(\cdot)$ to recover the original number of channels, followed by batch normalization $BN(\cdot)$. Finally, the normalized feature maps are added bit-wise to the initial input x .

The $MA(\cdot)$ can be formulated as in Equation 5:

$$MA(x) = x + BN(PWC_C(CSF(\sum DWC_{k_i} (R6(BN(PWC_{C \times 2}(x))))))) \quad (5)$$

E. Strip Convolution Block (SC)

Strip Convolution Block (SC)[28], [29], [30] is integrated into our SLMambaMedseg to address the limitations of traditional square-kernel convolutions in capturing elongated and curvilinear anatomical structures.

SC is designed as shown in Figure 1 (e). The input x first passes through a layer normalization, a fully connected layer, and a GELU activation function. The normalized patches x' then enter the Strip Module, which is the core part of SC depicted in Figure 1 (j).

In the Strip Module, a depthwise convolution with a large kernel (i.e., 5×5) is first applied to extract local contextual features. Then, two sequential depthwise convolutions with strip-shaped kernels (i.e., kernel size is $1 \times N$ and $N \times 1$) are used to better capture objects of high aspect ratios, followed by a pointwise convolution.

The output of the Strip Module, denoted as Z , undergoes an bit-wise multiplication (Hadamard product) with x' , yielding Z' , which then passes through an bit-wise addition with x after a fully connected layer. Subsequently, the fully connected output is added bit-wise to x again, producing Y . Next, the normalized Y is passed through a feed-forward network where features are refined and enhanced, followed by another bit-wise addition with the normalized Y , yielding the final output $SC(X)$.

F. Loss Function

To optimize the SLMambaMedseg for high-precision medical image segmentation, we employ a hybrid loss function, denoted as L_{total} , which synergistically combines Soft Dice Loss and Focal Loss.

The rationale behind this design is two-fold. First, Soft Dice Loss is utilized to maximize the spatial overlap between the segmented masks and the ground truth, effectively mitigating the issues caused by the severe class imbalance inherent in medical datasets (e.g., small

lesion areas versus large background regions). Second, to address the challenge of hard-to-classify pixels near anatomical boundaries, Focal Loss is integrated. By introducing a modulating factor $(1 - p_t)^\gamma$, Focal Loss reshapes the standard cross-entropy loss to down-weight easy examples and focus the model's learning capacity on ambiguous regions and sparse foreground pixels.

The objective function is formulated as in Equation 6:

$$L_{total} = \lambda_{dice} L_{Dice} + \lambda_{focal} L_{Focal} \quad (6)$$

where λ_{dice} and λ_{focal} are hyper-parameters representing the weighting coefficients for each component. In our experiments, we empirically set $\lambda_{dice} = 1$ and $\lambda_{focal} = 1$ to maintain a balanced gradient flow throughout the backpropagation process, ensuring both global structural consistency and local boundary sharpness.

IV. EXPERIMENTS

This section will present the details of implementation followed by a comparative analysis of our SLMambaMedseg against SOTA methods, including the performance of the complete model on the datasets and the details of the ablation study.

A. Implementation details

To evaluate the proposed SLMambaMedseg, comprehensive experiments are conducted on two widely adopted public benchmarks: the Synapse multi-organ segmentation dataset [31] and the Automated Cardiac Diagnosis Challenge (ACDC) dataset [32]. During pre-processing, all images are uniformly resized to a spatial resolution of 224×224 . To mitigate the risk of overfitting and enhance the generalization capability of the model, standard data augmentation techniques, including random flipping and random rotation, are employed. We set the batch size to 32 and employ AdamW optimizer [33] with the initial learning rate configured to 1×10^{-5} and weight decay configured to 1×10^{-3} . Furthermore, a cosine annealing learning rate scheduler [34] is applied to dynamically adjust the learning rate, utilizing a maximum of 50 iterations and a minimum learning rate of 1×10^{-5} . The entire training process spans 300 epochs. For quantitative assessment, two primary evaluation metrics are utilized: the Dice Similarity Coefficient (DSC) to measure region-based overlap [35], and the 95% Hausdorff Distance (HD95) to evaluate boundary-based accuracy [36]. All experiments are implemented using the PyTorch framework and executed on a single NVIDIA RTX 5090 GPU.

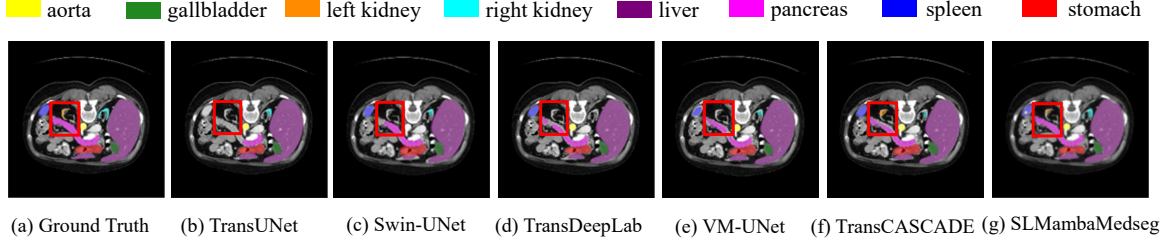


Fig. 2. Qualitative results on Synapse datasets are shown. The red rectangular box highlights incorrectly segmented polyps by SOTA methods.

Methods	DICE $\uparrow$	HD95 $\downarrow$	Aor.	Gal.	Kid.(L)	Kid.(R)	Liv.	Pan.	Spl.	Sto.
UNet[1]	70.11	44.69	84.00	56.70	72.41	62.64	86.98	48.73	81.48	67.96
AttnUNet[26]	71.70	34.47	82.61	61.94	76.07	70.42	87.54	46.70	80.67	67.66
TransUNet[4]	77.48	31.69	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
SSFormer[38]	78.01	25.72	82.78	63.74	80.72	78.11	93.53	61.53	87.07	76.61
PolypPVT[39]	78.08	25.61	82.34	66.14	81.21	73.78	94.37	59.34	88.05	79.4
Swin-UNet[5]	79.56	21.55	85.47	66.53	83.28	79.61	94.29	56.58	90.66	76.60
MT-UNet[40]	78.59	26.59	87.92	64.99	81.47	77.29	93.06	59.46	87.75	76.81
MEW-UNet[41]	78.92	16.44	86.68	65.32	82.87	80.02	93.63	58.36	90.19	74.26
PVT-CASCADE[42]	81.06	20.23	83.01	70.59	82.23	80.37	94.08	64.43	90.1	83.69
TransDeepLab[43]	80.16	21.25	86.04	69.16	84.08	79.88	93.53	61.19	89.00	78.40
VM-UNet[9]	81.08	19.21	86.40	69.41	86.16	82.76	94.17	58.80	89.51	81.40
TransCASCADE[37]	82.68	17.34	86.63	68.48	87.66	84.56	94.43	65.33	90.79	83.52
SLMambaMedseg(ours)	83.36	16.18	88.43	70.94	88.57	84.79	94.33	64.59	91.23	84.01

TABLE I

RESULTS ON SYNAPSE MULTI-ORGAN SEGMENTATION. DICE SCORES ARE REPORTED FOR INDIVIDUAL ORGANS. $\uparrow$ DENOTES HIGHER THE BETTER, $\downarrow$ DENOTES LOWER THE BETTER. THE BEST RESULTS ARE IN BOLD.

B. Experimental Results

SLMambaMedseg have compared with many state-of-the-art models with different type of architecture on Synapse and ACDC datasets. Pure CNN models like UNet[1]. Pure Transformer models like Swin-UNet[5], [5]. Pure Mamba models like VM-UNet[9]. Hybrid architecture models like TransUNet[4] that uses CNN and Transformer. And other SOTA models such as TransCASCADE[37] and so on. The later result details indicate our SLMambaMedseg achieves best among the SOTAs.

1) *Results on Synapse Multi-Organ Segmentation:* Results of experiments on Synapse are presented in Table I. Among SOTAs our SLMambaMedseg achieves the best performance, boasting a mean DICE score of 83.56 and a mean HD95 of 16.18, surpassing TransCASCADE[37] and VM-UNet[9]. There are two facts help our model, one is Mamba encoder costs less resources but predicts more precisely, the other one is our SLMambaMedseg employs large-kernel and strip convolution that successfully enhance capacity of feature extraction. Figure 2 shows visual results of some SOTA models on Synapse segmentation. Evidently, several SOTA models exhibit misclassifications. TransUNet[4], Swin-UNet[5], TransDeepLab[43] and VM-UNet[9] miss left kidney. TransUNet[4] and Swin-

UNet[5] even lost very big part of pancreas. A similar limitation is observed in TransDeepLab[43] but not serious like the former two. Our SLMambaMedseg performs best, especially in segmentation on tiny and slender organs such as both right and left kidney.

2) *Results of ACDC Segmentation:* Results of experiments on ACDC are presented in Table II. Our model achieves the best performance with a mean DICE score of 92.81. Figure 2 presents visual results of SOTA models performing on ACDC segmentation. SLMambaMedseg get a better performance than Swin-UNet[5], TransUNet[4] and TransDeepLab. Swin-UNet[5] predict incorrectly, missing the middle part of left ventricle. TransUNet[4] similarly but miss the left part. TransDeepLab[43] miss parts of Myo and captures wrong extra parts for left ventricle. Additionally, VM-UNet[9] and TransCASCADE[37] predict around correctly, however, not smoothly or precisely as our SLMambaMedseg.

C. Ablation Studies

1) *Effect of Mamba Encoder:* To understand the effect of different encoders, a set of experiments are conducted on the Synapse and ACDC datasets. Two extra version of SLMambaMedseg, SLCNNMedseg and SLTransMedseg by replacing the encoder using CNN(as UNet[1]) and Transformer(as SwinUNet[5]), have been

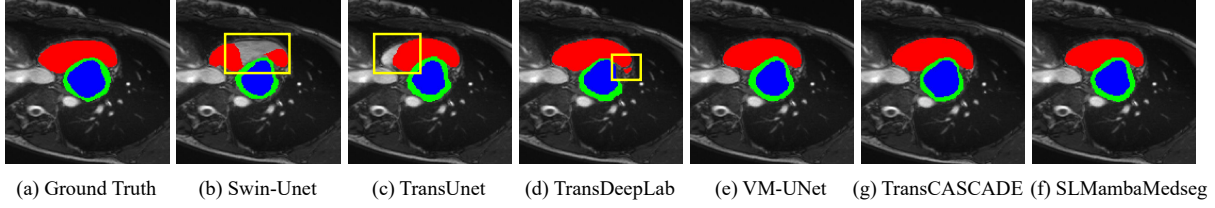


Fig. 3. Qualitative results on ACDC datasets are shown. Red represents the right ventricle, blue represents the left ventricle, and green represents the myocardium. The yellow rectangular box highlights incorrectly segmented polyps by SOTA methods.

Methods	DICE↑	RV	Myo	LV
TransUNet[4]	89.71	86.67	87.27	95.18
SwinUNet[5]	90.00	88.55	85.62	95.83
HiFormer[44]	90.12	91.06	84.54	94.77
TransDeepLab[43]	90.41	91.45	84.54	95.23
MT-UNet[40]	90.43	86.64	89.04	95.62
MISSFormer[45]	90.86	89.55	88.04	94.99
ParaTransCNN[46]	91.31	92.76	85.84	95.34
VM-UNet[9]	91.63	90.25	89.14	95.50
TransCASCADE[37]	92.12	90.65	89.68	96.02
SLMambaMedseg(ours)	92.81	91.52	90.58	96.53

TABLE II

RESULTS ON ACDC. DICE SCORES(%) ARE REPORTED FOR INDIVIDUAL ORGANS. ↑ DENOTES HIGHER THE BETTER. THE BEST RESULTS ARE IN BOLD.

Methods	Synapse		ACDC
	DICE↑	HD95↓	DICE↑
SLCNNMedseg	81.93	18.27	89.94
SLTransMedseg	82.76	16.62	91.05
SLMambaMedseg	83.36	16.18	92.81

TABLE III

RESULTS OF HOW DIFFERENT ENCODERS AFFECT PERFORMANCE OF SEGMENTATION ON SYNAPSE AND ACDC DATASETS. ↑ DENOTES HIGHER THE BETTER, ↓ DENOTES LOWER THE BETTER. THE BEST RESULTS ARE IN BOLD.

built up to be compared. Table III shows the experimental results. Our SLMambaMedseg with vision mamba encoder have achieved the best scores on both the Synapse and ACDC datasets, surpassing SLCNNMedseg and SLTransMedseg. Such results are completely in line with our expectations. The capability of CNNs in processing image features can no longer meet the surging demands, and their performance falls short when compared to Transformers and Mamba. However, Transformers also have limitations, as an excessive amount of data not only leads to a sharp increase in parameter counts but may also negatively impact segmentation results. In contrast, the Mamba encoder's ability to compress data in long-sequence[47] tasks is likely the key factor that enables it to outperform Transformers.

Methods	Synapse			ACDC
	DICE↑	HD95↓	Kid.(L)	DICE↑
No-SLGAG	81.80	18.47	83.97	90.79
SLMambaMedseg	83.36	16.18	88.57	92.81

TABLE IV

RESULTS OF ABLATION EXPERIMENT OF HOW STRIP LARGE-KERNEL GROUP ATTENTION GATE(SLGAG) AFFECTS ON SYNAPSE AND ACDC DATASETS. ↑ DENOTES HIGHER THE BETTER, ↓ DENOTES LOWER THE BETTER.

2) *Effect of SLGAG*: To observe the actual effectiveness of our SLGAG, one specific experiment has been designed by removing the skip connections composed of SLGAG from SLMambaMedseg and use unprocessed skip connections instead, comparing this variant with our existing SLMambaMedseg on the Synapse and ACDC datasets. The experimental results are shown in Table IV. The SLMambaMedseg with SLGAG achieves higher scores than the model without SLGAG, particularly demonstrating a significant lead in segmentation tasks involving small and elongated structures, such as the left kidney on the Synapse dataset. This indicates that the strip convolutions in our SLGAG can effectively capture relatively small and elongated features.

3) *Effect of Convolution Kernel Size in MA*: A series of experiments have been designed to evaluate the performance of the SLCB module. Specifically, experiments on the Synapse and ACDC datasets are conducted, with variable number and size of depthwise convolution kernels within the MA module. There are several strategies, including the use of a single 3×3 kernel, multiple 3×3 kernels, multiple 5×5 kernels, and a combination of multi-scale kernels. The experimental results are shown in Table V. Ultimately, our tests demonstrate that the integrated use of three kernels with sizes 1, 3, and 5 achieves the optimal segmentation results. It is hypothesized that such multi-scale depthwise convolutions extract local and global features as uniformly as possible, thereby enhancing the efficiency of the process. Notably, increasing the kernel size may lead to a decline in segmentation performance; this is

Methods	Kernel Strategies									
	[1]	[3]	[5]	[1,3]	[3,5]	[3,3,3]	[1,3,5]	[5,5,5]	[3,5,7]	[1,3,5,7]
Synapse	83.11	82.93	82.64	83.02	83.25	82.73	83.36	82.98	83.15	83.20
ACDC	91.14	91.94	91.89	92.28	91.93	92.35	92.81	92.32	92.12	92.43

TABLE V

RESULTS OF DIFFERENT DEPTHWISE CONVOLUTION KERNEL STRATEGIES EFFECT ON SYNAPSE AND ACDC. IN $[k_1, k_2, \dots, k_n]$, k_i MEANS SIZE OF EACH KERNEL, $\sum_1^n k_i$ MEANS TOTAL NUMBER OF THE KERNELS. THE BEST RESULTS ARE IN BOLD.

likely because excessively large kernels overlook local features, resulting in suboptimal performance in tasks that emphasize the segmentation of small objects.

V. CONCLUSIONS

This paper propose SLMambaMedseg, a novel and efficient medical image segmentation framework built upon Vision Mamba and Multi-Scale Large-kernel Strip Convolutions. The model is designed to enhance segmentation efficiency while meeting the rigorous demands of real-world clinical applications. By synergizing the high-performance Vision Mamba backbone with multi-scale large-kernel and strip convolutions, SLMambaMedseg significantly improves the precision of both global context modeling and local feature extraction, demonstrating superior efficiency particularly in delineating small, elongated, and intricate anatomical structures.

Experimental results on the Synapse and ACDC datasets indicate that SLMambaMedseg outperforms state-of-the-art models, including VM-UNet and TransCASCADE, in terms of mean segmentation accuracy. Furthermore, extensive ablation studies validate the feasibility and potential of integrating large-kernel and strip convolutions into medical imaging tasks. Our findings suggest that Vision Mamba is exceptionally well-suited for constructing robust encoding modules, while the combination of large-kernel and strip convolutions facilitates the design of lightweight yet high-efficiency decoders. We posit that Vision Mamba will increasingly supersede Transformers in future especially in lightweight scenarios; when coupled with emerging convolutional techniques, it is poised to further elevate the overall performance of diverse computer vision tasks.

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